

K8s-03

1. Create a ClusterIP service for an Apache web server pod.

- Create Apache pod file

```
root@Master:~# mkdir -p ~/k8s-03
cd ~/k8s-03
root@Master:~/k8s-03# vi apache-pod.yaml
root@Master:~/k8s-03#
```

```
apiVersion: v1
kind: Pod
metadata:
  name: apache-pod
  labels:
    app: apache
spec:
  containers:
  - name: apache
    image: httpd
    ports:
    - containerPort: 80
```

- Create pod and check pods status

```
root@Master:~/k8s-03# kubectl apply -f apache-pod.yaml
pod/apache-pod created
root@Master:~/k8s-03# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
apache-pod    1/1     Running   0           11s
createpod     1/1     Running   2 (7m30s ago)  18h
```

- Create apache-service. Yaml

```
apiVersion: v1
kind: Service
metadata:
  name: apache-service
spec:
  selector:
    app: apache
  ports:
  - port: 80
    targetPort: 80
  type: ClusterIP
```

- Apply the Service

```
root@Master:~/k8s-03# vi apache-service.yaml
root@Master:~/k8s-03# kubectl apply -f apache-service.yaml
service/apache-service created
root@Master:~/k8s-03#
```

- Verify Service and Test the Service

```
root@Master:~/k8s-03# kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
apache-service      ClusterIP   10.102.172.16 <none>       80/TCP     63s
firstpod-service    ClusterIP   10.99.160.6    <none>       80/TCP     5h23m
kubernetes           ClusterIP   10.96.0.1      <none>       443/TCP    25h
root@Master:~/k8s-03# kubectl get pod apache-pod -o wide
NAME                READY    STATUS    RESTARTS   AGE    IP              NODE    NOMINATED NODE    READINESS GATES
apache-pod           1/1      Running   0           5m40s  192.168.219.105 master   <none>
```

- Check endpoints:

```
root@Master:~/k8s-03# kubectl get endpoints apache-service
NAME                ENDPOINTS                AGE
apache-service      192.168.219.105:80      2m27s
root@Master:~/k8s-03#
```

2. Expose a Nginx pod externally using a NodePort service.

- Create a nginx-nodeport file

```
root@Master:~/k8s-03# vi nginx-nodeport-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-nodeport
  labels:
    app: nginx-nodeport
spec:
  containers:
  - name: nginx
    image: nginx
    ports:
    - containerPort: 80
```

- Apply

```
root@Master:~/k8s-03# kubectl apply -f nginx-nodeport-pod.yaml
pod/nginx-nodeport created
root@Master:~/k8s-03#
```

- Create nginx-nodeport-service. Yaml

```
root@Master:~/k8s-03# vi nginx-nodeport-service.yaml
root@Master:~/k8s-03#
apiVersion: v1
kind: Service
metadata:
  name: nginx-nodeport-service
spec:
  selector:
    app: nginx-nodeport
  type: NodePort
  ports:
  - port: 80
    targetPort: 80
    nodePort: 30080
```

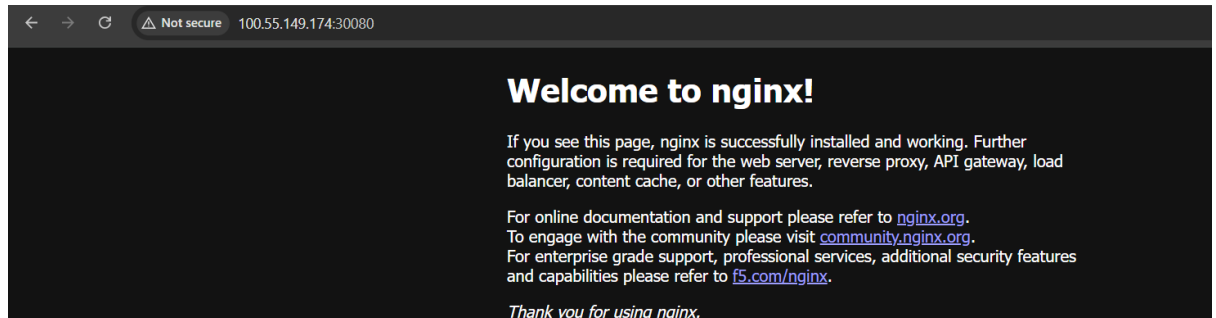
- Apply it and verify svc

```
root@Master:~/k8s-03# kubectl apply -f nginx-nodeport-service.yaml
service/nginx-nodeport-service created
root@Master:~/k8s-03# kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
apache-service	ClusterIP	10.102.172.16	<none>	80/TCP	17m
firstpod-service	ClusterIP	10.99.160.6	<none>	80/TCP	5h39m
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	25h
nginx-nodeport-service	NodePort	10.110.94.8	<none>	80:30080/TCP	6s

```
root@Master:~/k8s-03#
```

- Open in browser with master public Ip and port no 30080



3. Deploy a Replication Controller to maintain 3 replicas of a Nginx pod.

- Create nginx-rc.yaml

```
root@Master:~/k8s-03# vi nginx-rc.yaml
root@Master:~/k8s-03#
```

```
apiVersion: v1
kind: ReplicationController
metadata:
  name: nginx-rc
spec:
  replicas: 3
  selector:
    app: nginx-rc
  template:
    metadata:
      labels:
        app: nginx-rc
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

- Apply it and verify the RC

```
root@Master:~/k8s-03# kubectl apply -f nginx-rc.yaml
replicationcontroller/nginx-rc created
root@Master:~/k8s-03# kubectl get rc
```

NAME	DESIRED	CURRENT	READY	AGE
nginx-rc	3	3	3	7s

```
root@Master:~/k8s-03#
```

- Verify the Pods

```
root@Master:~/k8s-03# kubectl get pods -l app=nginx-rc
NAME                READY   STATUS    RESTARTS   AGE
nginx-rc-7pxvv      1/1     Running   0           53s
nginx-rc-j5gsm      1/1     Running   0           53s
nginx-rc-ssfpb      1/1     Running   0           53s
root@Master:~/k8s-03# kubectl describe pod nginx-rc-7pxvv
```

- When I tried delete one pod it after that it automatically creates new

```
root@Master:~/k8s-03# kubectl delete pod nginx-rc-7pxvv
pod "nginx-rc-7pxvv" deleted
root@Master:~/k8s-03# kubectl get pods -w
NAME                READY   STATUS    RESTARTS   AGE
apache-pod          1/1     Running   0           39m
nginx                1/1     Running   2 (46m ago) 19h
nginx-nodeport      1/1     Running   0           23m
nginx-rc-j5gsm      1/1     Running   0           7m54s
nginx-rc-pxrmg      1/1     Running   0           9s
nginx-rc-ssfpb      1/1     Running   0           7m54s
^Croot@Master:~/k8s-03#
```

4. Scale the Replication Controller from 3 replicas to 5 replicas.

- Open RC file and change replica 3 to 5 and applied

```
root@Master:~/k8s-03# vi nginx-rc.yaml
root@Master:~/k8s-03# vi nginx-rc.yaml
root@Master:~/k8s-03# kubectl apply -f nginx-rc.yaml
replicationcontroller/nginx-rc configured
```

```
apiVersion: v1
kind: ReplicationController
metadata:
  name: nginx-rc
spec:
  replicas: 5
  selector:
    app: nginx-rc
  template:
    metadata:
      labels:
        app: nginx-rc
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

- Now verify the pod and check the pods

```
root@Master:~/k8s-03# kubectl get rc
NAME      DESIRED   CURRENT   READY   AGE
nginx-rc  5         5         5       15m
root@Master:~/k8s-03# kubectl get pods -l app=nginx-rc
NAME                READY   STATUS    RESTARTS   AGE
nginx-rc-8mqgz      1/1     Running   0           2m17s
nginx-rc-j5gsm      1/1     Running   0           15m
nginx-rc-pxrmg      1/1     Running   0           7m25s
nginx-rc-ssfpb      1/1     Running   0           15m
nginx-rc-z45v9      1/1     Running   0           2m16s
root@Master:~/k8s-03#
```

5. Create a ReplicaSet to manage pods based on multiple labels (prod and test).

- Create multilabel-rs. Yaml

```
root@Master:~/k8s-03# vi multilabel-rs.yaml
root@Master:~/k8s-03#
```

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: multilabel-rs
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
      env: prod
      team: test
  template:
    metadata:
      labels:
        app: nginx
        env: prod
        team: test
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

- Apply and Verify ReplicaSet

```
root@Master:~/k8s-03# kubectl apply -f multilabel-rs.yaml
replicaset.apps/multilabel-rs created
root@Master:~/k8s-03# kubectl get rs
NAME           DESIRED   CURRENT   READY   AGE
multilabel-rs   3         3         3       29s
```

- Verify Pods and labels

```
root@Master:~/k8s-03# kubectl get pods -l app=nginx,env=prod,team=test
NAME                READY   STATUS    RESTARTS   AGE
multilabel-rs-bv9mk  1/1     Running   0          39s
multilabel-rs-xm26v  1/1     Running   0          39s
multilabel-rs-zxr4p  1/1     Running   0          39s
root@Master:~/k8s-03# kubectl get pods --show-labels | grep multilabel-rs
multilabel-rs-bv9mk  1/1     Running   0          50s   app=nginx,env=prod,team=test
multilabel-rs-xm26v  1/1     Running   0          50s   app=nginx,env=prod,team=test
multilabel-rs-zxr4p  1/1     Running   0          50s   app=nginx,env=prod,team=test
root@Master:~/k8s-03#
```

6. Deploy a ReplicaSet that excludes pods with the label backend.

- Create exclude-backend-rs. yaml

```
root@Master:~/k8s-03# vi exclude-backend-rs.yaml
root@Master:~/k8s-03#
```

```

apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: exclude-backend-rs
spec:
  replicas: 2
  selector:
    matchExpressions:
      - key: backend
        operator: DoesNotExist
      - key: app
        operator: In
        values:
          - frontend
  template:
    metadata:
      labels:
        app: frontend
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80

```

- Appl and Verify ReplicaSet

```

root@Master:~/k8s-03# kubectl apply -f exclude-backend-rs.yaml
replicaset.apps/exclude-backend-rs created
root@Master:~/k8s-03# kubectl get rs
NAME                DESIRED   CURRENT   READY   AGE
exclude-backend-rs  2         2         2       8s
multilabel-rs       3         3         3       6m10s
root@Master:~/k8s-03#

```

- Verify pods and labels

```

root@Master:~/k8s-03# kubectl get pods -l app=frontend
NAME                                READY   STATUS    RESTARTS   AGE
exclude-backend-rs-22jq6            1/1     Running   0           77s
exclude-backend-rs-pkqmg            1/1     Running   0           77s
root@Master:~/k8s-03# kubectl get pods --show-labels | grep frontend
exclude-backend-rs-22jq6            1/1     Running   0           85s    app=frontend
exclude-backend-rs-pkqmg            1/1     Running   0           85s    app=frontend
root@Master:~/k8s-03#

```

7. Test load balancing across multiple pods using a NodePort service.

- Create rc-nodeport-service. Yaml

```

root@Master:~/k8s-03# vi rc-nodeport-service.yaml
root@Master:~/k8s-03#

```

```

apiVersion: v1
kind: Service
metadata:
  name: nginx-rc-service
spec:
  selector:
    app: nginx-rc
  type: NodePort
  ports:
    - port: 80
      targetPort: 80
      nodePort: 3008

```

- Apply and verify it

```
root@Master:~/k8s-03# kubectl apply -f rc-nodeport-service.yaml
service/nginx-rc-service created
root@Master:~/k8s-03# kubectl get svc
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
apache-service	ClusterIP	10.102.172.16	<none>	80/TCP	70m
firstpod-service	ClusterIP	10.99.160.6	<none>	80/TCP	6h32m
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	26h
nginx-nodeport-service	NodePort	10.110.94.8	<none>	80:30080/TCP	53m
nginx-rc-service	NodePort	10.103.111.115	<none>	80:30081/TCP	8s

```
root@Master:~/k8s-03#
```

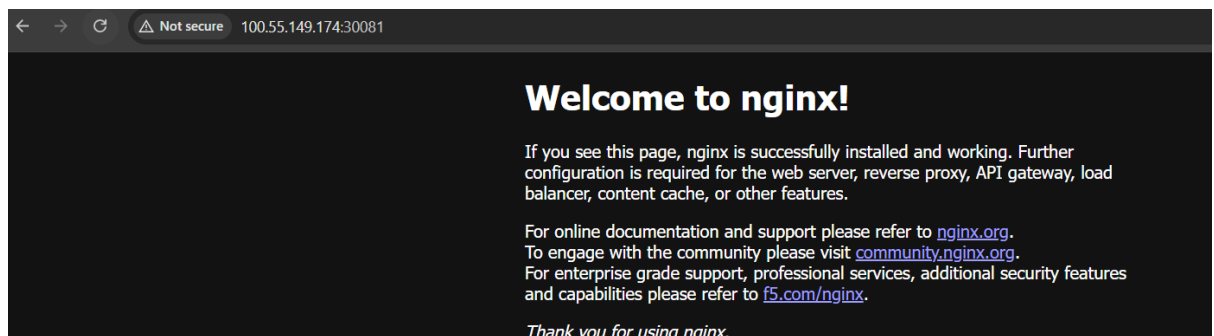
- Verify endpoints

```
root@Master:~/k8s-03# kubectl get endpoints nginx-rc-service
```

NAME	ENDPOINTS	AGE
nginx-rc-service	192.168.219.108:80,192.168.219.109:80,192.168.219.110:80 + 2 more...	2m8s

```
root@Master:~/k8s-03#
```

- Open browser check with master IP and port no 30081



8. Delete a Replication Controller without affecting the running pods.

- Delete RC (Replication Controller deleted)

```
root@Master:~/k8s-03# kubectl delete rc nginx-rc --cascade=orphan
r AWS Console Home roller "nginx-rc" deleted
```

- Now verify RC (Replication Controller no longer exists)

```
root@Master:~/k8s-03# kubectl get rc
No resources found in default namespace.
```

- Now verify pods are running or not (Pods are still running)

```
root@Master:~/k8s-03# kubectl get pods -l app=nginx-rc
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-rc-8mqgz	1/1	Running	0	40m
nginx-rc-j5gsm	1/1	Running	0	53m
nginx-rc-pxrmg	1/1	Running	0	45m
nginx-rc-ssfpb	1/1	Running	0	53m
nginx-rc-z45v9	1/1	Running	0	40m

```
root@Master:~/k8s-03#
```